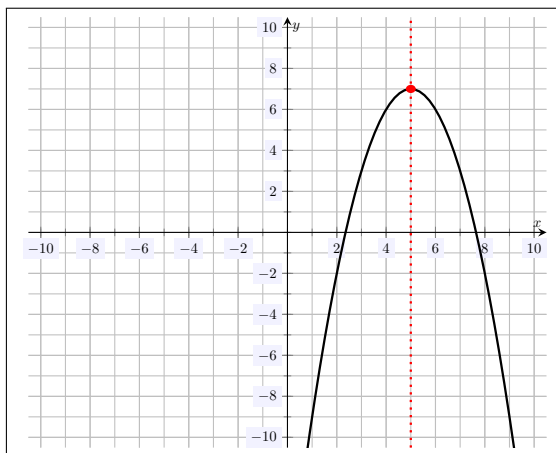


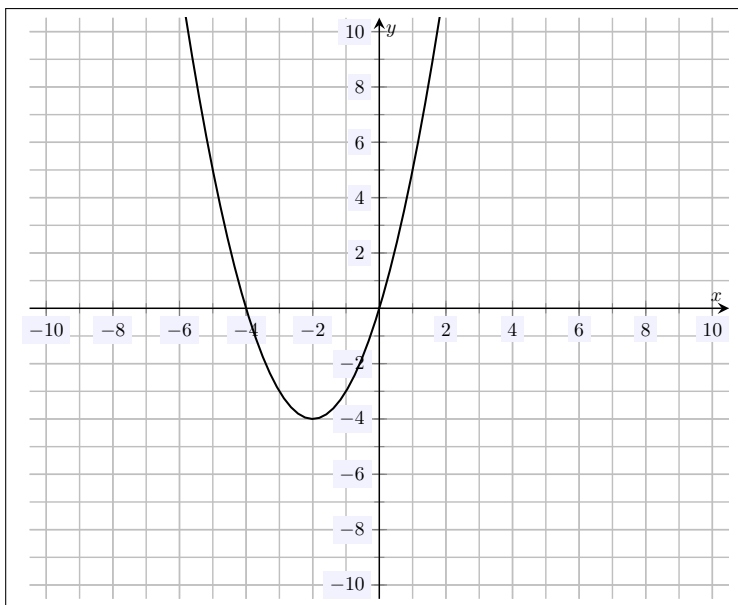
Recitation: Polynomials (Linear & Quadratic Functions) Solutions

Problem 1. Sketch the function $f(x) = 7 - (x - 5)^2$.



We know $f(x)$ has vertex $(5, 7)$, axis of symmetry $x = 5$, and opens downwards (because $a = -1 < 0$). Using this information gives us the sketch above.

Problem 2. Find the equation of the quadratic function shown below. Be sure to fully justify why your answer is correct.



By the Fundamental Theorem of Algebra, we know the quadratic above is $a(x - r_1)(x - r_2)$, where the r_i are the roots. We see this quadratic has roots $x = -4, 0$. So, the quadratic is $a(x - (-4))(x - 0) = ax(x + 4)$. But we see this parabola also contains the points $(-5, 5), (-2, -4), (1, 5)$. Using, for example, $(1, 5)$, we know $y = 5$ when $x = 1$, but then $5 = a \cdot 1(1 + 4)$, i.e. $5 = 5a$. Therefore, $a = 1$. So, the quadratic is $x(x + 4)$.

Alternatively, we know the quadratic has the (vertex) form $a(x - P)^2 + Q$, where (P, Q) is the vertex. We see this has vertex $(-2, -4)$. So, the quadratic is $a(x - (-2))^2 - 4 = a(x + 2)^2 - 4$. But we see this parabola also contains the points $(-5, 5), (-4, 0), (0, 0), (1, 5)$. Using, for example, $(0, 0)$, we know $y = 0$ when $x = 0$. But then $0 = a(0 + 2)^2 - 4$, so that $4a = 4$, i.e. $a = 1$. Therefore, the quadratic is $(x + 2)^2 - 4$.

Problem 3. Solve the following equation:

$$5\sqrt{2}x + 8 = -3(1 - 2x)$$

$$5\sqrt{2}x + 8 = -3(1 - 2x)$$

$$5\sqrt{2}x + 8 = -3 + 6x$$

$$5\sqrt{2}x - 6x = -11$$

$$(5\sqrt{2} - 6)x = -11$$

$$x = \frac{-11}{5\sqrt{2} - 6}$$

Problem 4. Consider the quadratic function $f(x) = x^2 - 6x + 14$.

- (a) Find a, b, c for this quadratic function.
- (b) Does $f(x)$ open upwards or downwards? Explain.
- (c) Is this quadratic function convex or concave? Explain.
- (d) Find the minimum value of $f(x)$, if it exists. If it does not exist, explain why.
- (e) Find the maximum value of $f(x)$, if it exists. If it does not exist, explain why.

Solution.

- (a) We have $a = 1$, $b = -6$, $c = 14$.
- (b) Because $a = 1 > 0$, the quadratic function opens upwards.
- (c) Because $a = 1 > 0$, the quadratic function is convex.
- (d) Because $a = 1 > 0$, the quadratic function has a minimum value—namely, the y -coordinate of the vertex. Completing the square, we have $x^2 - 6x + 14 = x^2 - 6x + 9 - 9 + 14 = (x - 3)^2 + 5$. So, the minimum possible output is 5. Alternatively, using the evaluation method, we know the minimum occurs at $x = -\frac{b}{2a} = -\frac{-6}{2(1)} = 3$. Then $f(3) = 3^2 - 6(3) + 14 = 9 - 18 + 14 = 5$.
- (e) Because $a = 1 > 0$, the quadratic function has no maximum value.